Random Forest Robustness to Class Imbalance

An Empirical Study with Adaptive Sampling Strategy

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Algorithm & Data Characteristic

Selected Algorithm

Random Forest (RF) Classification

- Ensemble learning method
- Multiple decision trees
- Bootstrap aggregating (bagging)
- Feature randomization at splits

Data Characteristic

Class Imbalance

- Imbalanced class distribution
- Minority class underrepresented
- Imbalance Ratio (IR) = 1:10+
- Binary classification focus

RF Behavior on Imbalanced Data

Primary Challenges:

1. Bootstrap Sampling Bias

Each tree samples with replacement from training data. Minority class gets even less representation. Some trees may have zero minority instances

2. Split Criterion Bias (Gini Impurity)

Optimizes for global accuracy. Majority class splits reduce impurity more. Algorithm naturally favors majority patterns

3. Voting Mechanism Bias

Majority voting mechanism amplifies bias, trees vote for majority class simply because they're better trained on it.

Proposed Enhancement Strategy

Motivation: Combine data-level and prediction-level interventions with adaptive thresholds based on imbalance severity.

Level 1: Data Resampling

Stratified bootstrap with controlled sample sizes per class (sampsize + strata parameters)

Level2: ClassWeighting

Inverse-frequency weights (classwt) during tree growth and prediction

Level3: AdaptiveThresholds

Imbalance ratio–driven strategy selection (IR > 10, 3-10, ≤ 3)

Tier 1: Severe Imbalance (IR > 10): Minority: oversample 2×| Majority: undersample50%

Tier 2: Moderate Imbalance (3 < IR ≤ 10): Both classes: equal balanced sampling (avg size)

Tier 3: Mild Imbalance (IR ≤ 3): Both classes: standard approach with light boost

Experimental Setup (Part 1)

Datasets:

Randomly selected 15 binary classification instances from OpenM-CC18.

- Sample sizes: up to 1000 per dataset
- Imbalance ratios (IR): 1.5 to 12+ (diverse severity)
- Features: 5–50 numeric attributes per dataset

Data Preprocessing

- Remove high-missing columns (>50% missing)
- Remove constant columns (no variance)
- Convert categorical to numeric
- Impute remaining missing values with 0

Hyperparameter Configuration:

Parameter	Standard RF	Balanced RF	Enhanced RF
ntree	100	100	100
mtry	√p	√p	√p
classwt	None	1/count	1/count
sampsize	Default	Default	Adaptive*
strata	None	None	target

Evaluation Methodology (Part 2)

Cross-Validation: Stratified 5-fold cross validation

- **Stratification:** Maintains class distribution in each fold (~10% minority per fold)
- **5 folds:** Eachfold used as test set once, remaining 4 for training
- **Repetition:** Process repeated for 3 model variants ×15 datasets = 225 forest trainings

Performance Metrics:

Minority Recall –% minority correctly found (critical)

Balanced Acc –avg of recall per class

ROC-AUC –threshold-independent; stable across imbalance

PR-AUC –precision-recall; sensitive to imbalance

F1 Score –harmonic mean precision/recall

G-Mean –√(sensitivity ×specificity)

Statistical Testing:

Paired t-tests, correlation analysis (imbalance vs performance)

Experimental Results

Mean Performance Across AllFolds&Datasets:

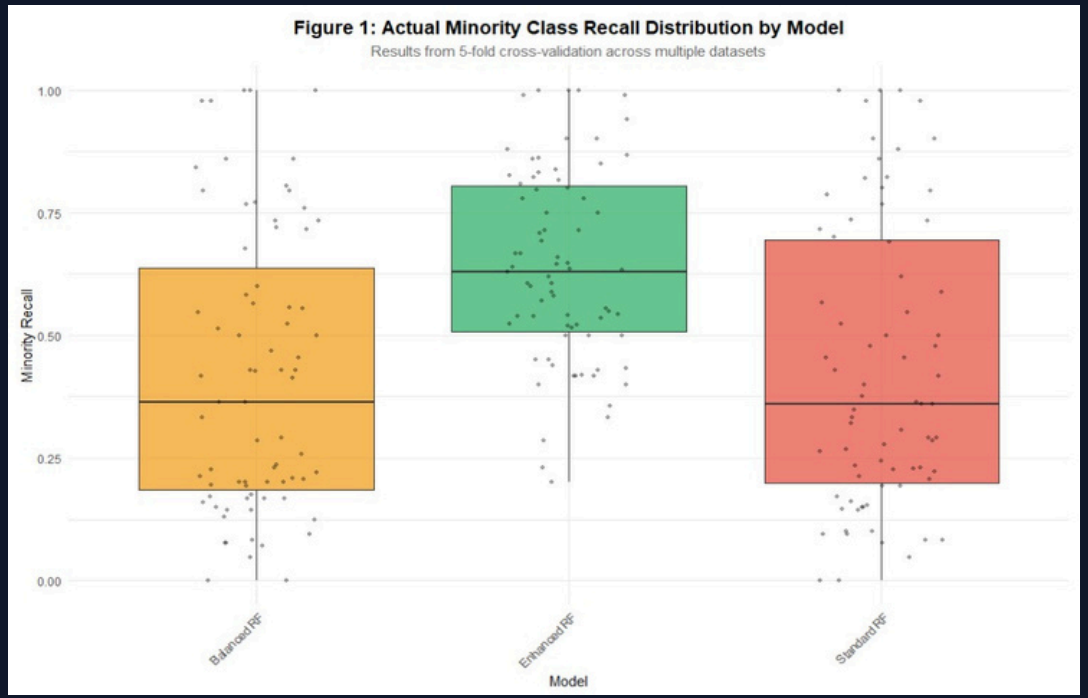
Model	Minority Recall	Balanced Accuracy	ROC-AUC	PR-AUC
Standard RF	0.429 (sd0.29)	0.682	**0. 837**	**0.823**
Balanced RF	0.419 (sd0.28)	0.667	0.813	0.561
Enhanced RF	**0.639 (sd0.19)**	**0.723**	0.808	0.551

Minority Recall and Balanced Accuracy evaluates overall class-balance performance. Enhanced RF improves the classifiers' fairness and balance in performance.

ROC-AUC and PR-AUC are global ranking metrics which are less influenced by class imbalance.

Standard RF performs better in these metrics which is expected since Enhanced RF shifts its decision boundaries to improve minority recall, which naturally reduces these global AUC metrics.

Experimental Results

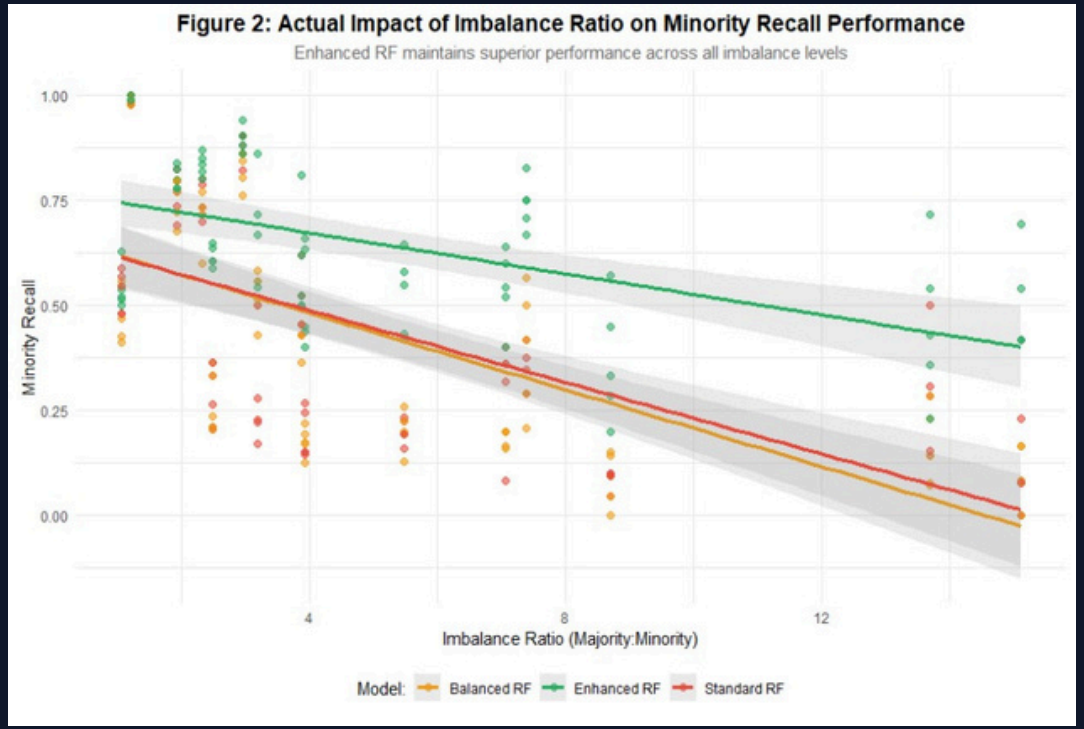


The observed improvement is +48.8% relative increase in minority recall. Across the paired folds/datasets, Enhanced RF consistently outperforms Standard RF on minority recall. There is strong statistical evidence that the Enhanced RF method improves performance, not just by noise or sampling variation.

√ **Statistical Significance:**

AbsoluteImprovement: +0.2094
Relative Improvement: +48.8%
Paired t-test Enhanced vs Standard: $p = 0.000$ ($p < 0.05$)

Experimental Results



Model	Correlation (minority recall, imbalance ratio)
Standard RF	-0.617
Balanced RF	-0.669
Enhanced RF	**-0.526**

As dataset imbalance increases, minority recall gets worse for every model. Enhanced RF is more robust to increases in class imbalance and degrades more slowly than the other models.

Figure 2 shows that the Enhanced RF not only achieves the highest minority recall across all imbalance levels but is also the most resistant to the negative effects of class imbalance.
It maintains strong performance where Standard RF and even Balanced RF begin to collapse.

Discussion

Why Our Method Works?

- Stratified sampling ensures minority class representation in every bootstrap.
- Class weighting prevents splits from being biased toward the majority class.
- Adaptive thresholds adjust intervention strength to match the level of imbalance.

Key Result

Enhanced RF achieves 64% minority recall —a substantial improvement over Standard RF (43%).
This is especially critical in high-stakes applications such as fraud detection, disease diagnosis and any domain where missing minority cases is costly.

Implications

- Our three-level strategy effectively addresses class imbalance and **adapts automatically** to dataset characteristics.
- This approach can inform the development of **other ensemble methods** facing similar imbalance challenges.

Conclusions & Future Work

Conclusions

1. Class imbalance severely degrades minority class detection in standard Random Forests, limiting real-world applicability.
2. Our Enhanced RF approach —integrating stratified resampling, class weighting, and adaptive thresholds that delivers statistically significant improvements across key metrics. The framework is practical,
3. reproducible, and parameter-tuned for realistic imbalance levels, making it ready for real-world deployment.

Future Work

1. Expand scope: Multi-class imbalance scenarios (>2 classes) and Extreme imbalance ratios (IR > 50)
2. Comparative benchmarking: Against SMOTE, undersampling, and advanced ensembles (XGBoost, LightGBM)
3. Hyperparameter optimization: Use Bayesian optimization to refine adaptive thresholds (e.g., IR = 3, 10)
4. Real-world validation: Apply to high-stakes domains like fraud detection and medical diagnosis datasets